

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

CHEMISTRY
PAPER 2 Theory

4024/2

NOVEMBER 2021 SESSION

2 hours

Additional materials:
Mathematical tables/ Electronic calculator (optional)

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on all separate answer paper used.

Section A

Answer all questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer any **four** questions.

Write your answers on the separate answer paper provided

At the end of the examination, fasten any separate answer paper used securely to the question paper.

Enter the numbers of **Section B** questions you have answered in the grid.

All essential working must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

A copy of the Periodic Table is printed on page 17.

FOR EXAMINER'S USE

Section A

Section B

TOTAL

This question paper consists of 17 printed pages and 3 blank pages.

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Section A

Answer all questions from this section.

Write your answers in the spaces provided on the question paper.

1. (a) Electronic configurations of elements **M**, **N** and **P** are shown.

M	2,8,2
N	2,8,6
P	2,8,8

- (i) Write the formula of the compound formed when **M** reacts with **N**.

- (ii) Name the type of bond formed when **M** reacts with **N**.

- (iii) Compare the reactivities of **M** and **P**.

- (b) (i) Complete Table 1.1 by stating whether the given process is an exothermic or an endothermic process.

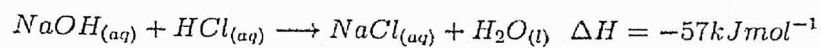
Table 1.1

process	type of process
formation of nitrogen from nitrogen molecules	
formation of a covalent bond	

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[2]

- (ii) Sodium hydroxide reacts with dilute hydrochloric acid according to the equation.



Draw an energy level diagram for the reaction.

[2]

2. (a) Fig. 2.1 shows apparatus set up to study the effect of copper on the reaction between zinc granules and sulphuric acid.

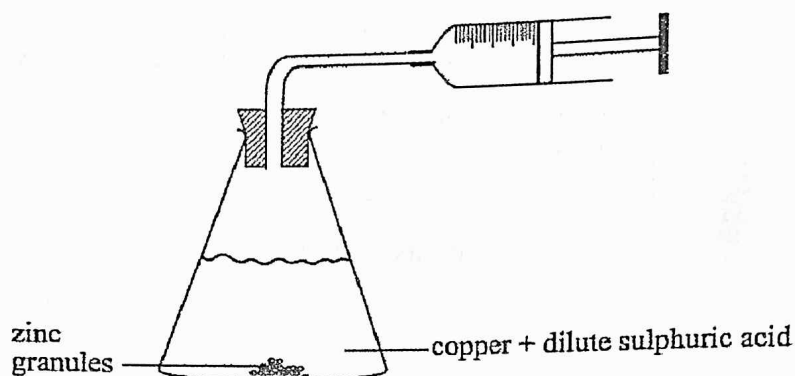


Fig.2.1

The experiment was repeated without copper.
Results obtained were recorded in Table 2.1

Table 2.1

experiment	reaction mixture	time taken for all zinc to react
1	zinc + dilute sulphuric acid + copper	121 seconds
2	zinc + dilute sulphuric acid	548 seconds

- (i) Explain the term *catalyst*.

[1]

- (ii) Write a balanced chemical equation for the reaction that occurred in the flask.

[2]

- (iii) Explain the difference in the observations.

.....

.....

.....

.....

.....

.....

- (b) (i) Name any other **two** factors which can affect the rate of the reaction. [2]

1.

2.

[2]

- (ii) Explain why most food preparation processes involves heating the food.

.....

.....

.....

[1]

3. (a) Structural formulae of some organic compounds are shown in Fig. 3.1

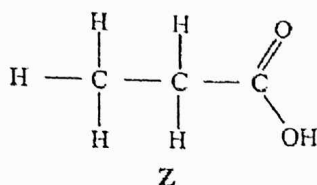
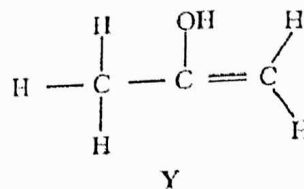
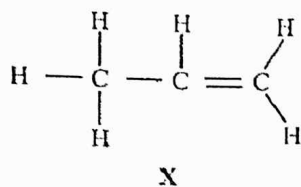


Fig. 3.1

- (i) Name the homologous series to which compound X belongs.

[1]

- (ii) Draw structural formula of the organic product formed when X reacts with bromine.

[1]

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- (iii) Draw structural formula of the organic product formed when two molecules of X are reacted.

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[1]

- (b) Describe a chemical test that can be used to distinguish between:

- (i) X and Y,

.....
.....

[1]

- (ii) X and Z.

.....
.....

[1]

- (c) State, with reasons, which compound X, Y or Z can be used as an antiseptic.

compound

reasons

[3]

4. (a) Fig. 4.1 shows a water purification plant.

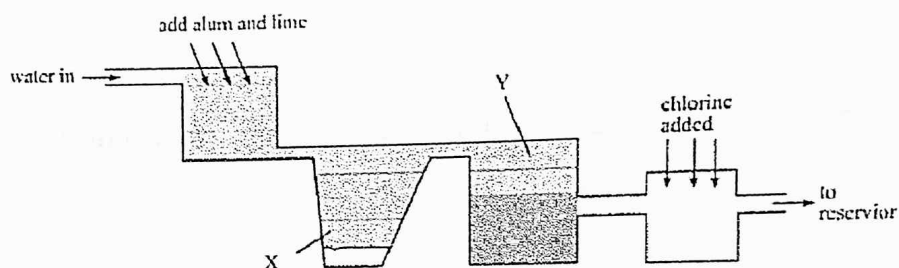


Fig. 4.1

- (i) Name the tanks labelled

1. X,

2. Y.

- (ii) Describe what happens in

1. X,

2. Y.

[2]

[2]

- (b) Describe and explain the importance of treating water for domestic use.

[2]

- (c) State any **two** differences in composition between water from a borehole and water from a flowing river.

1.

2.

[2]

5. (a) A mass of 2.5 g of a hydrocarbon, C_6H_{14} , was completely burnt.

- (i) Explain the term *complete combustion*.

[1]

- (ii) Calculate the volume of the gas produced in the reaction.

[2]

(b) (i) Describe how ethanol is manufactured from

1. ethene,

[2]

(ii) 2. carbohydrates.

[3]

Section B

Answer any **four** questions from this section.

Write your answers on a separate answer paper provided.

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6. (a) Fig 6.1 shows the structures of three atoms represented by the symbols X, Y and Z.

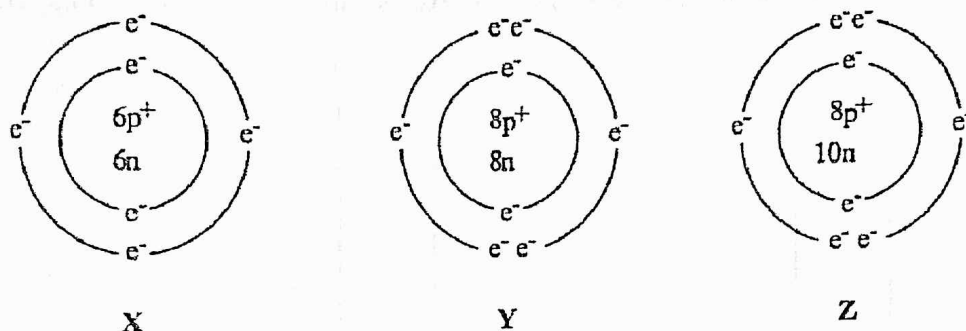


Fig. 6.1

- (i) Name the particle represented by e^- . [1]
- (ii) Describe and explain how atom Y and Z are related. [2]
- (iii) Write the nuclide notation of Z. [2]
- (b) Ammonium chloride decomposes when heated in a test tube as shown in Fig. 6.2

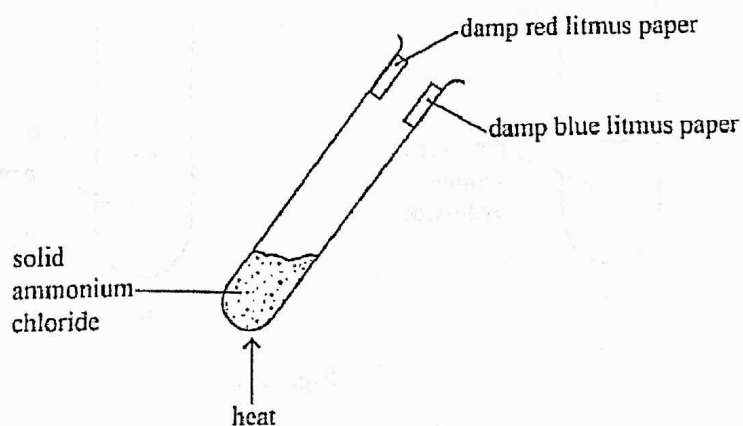


Fig. 6.2

- (i) Write an equation for the decomposition of ammonium chloride. [2]
- (ii) Describe and explain the observable changes that occur to the litmus papers. [6]
- (c) Describe the bonding in copper. [2]
7. (a) Four different chemicals were added to excess nitric acid as shown in Fig. 7.1.

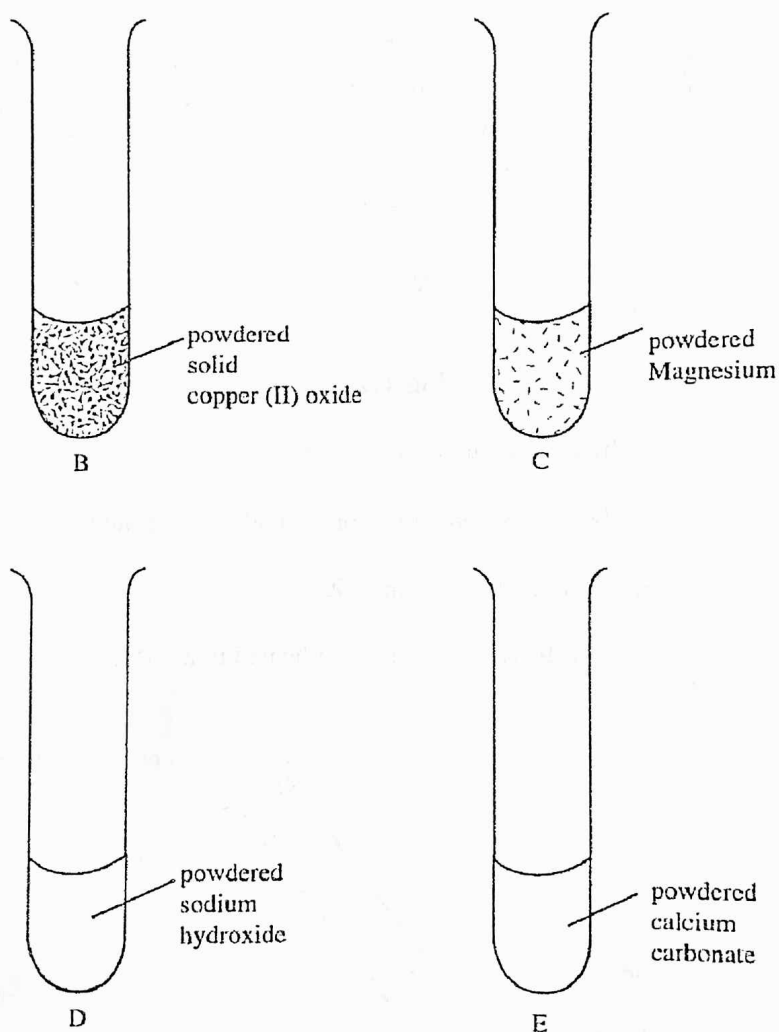


Fig. 7.1

- (i) Describe the observable changes that occur in test tube
1. B,
 2. E.
- [4]
- (ii) State with a reason the type of reaction occurring in test tube D. [2]
- (iii) Write an equation for the reaction occurring in test tube C. [2]
- (b) When iron and steam are strongly heated together in a closed container, a dynamic equilibrium is reached. The chemical equation for the reaction is shown:
- $$3Fe_{(s)} + 4H_2O_{(g)} \rightleftharpoons Fe_3O_{4(s)} + 4H_{2(g)} \quad \Delta H = -149 \text{ kJ/mol}$$
- (i) Define the term *dynamic equilibrium*? [1]
- (ii) Describe and explain any **three** ways of increasing the the percentage yield of $Fe_3O_{4(s)}$. [6]
8. (a) (i) Draw a dot and cross diagram to show the bonding in:
1. C_2H_4 ,
 2. CaF_2 .
- [2]
- (ii) Name the type of bonding in each of the compounds in (i). [2]
- (b) Describe and explain how the type of bonding affects electrical conductivity of
1. metallic solids,
 2. covalent compounds,
 3. ionic compounds.
- [6]

- (c) Fig. 8.1 shows some of the reactions undergone by limestone.

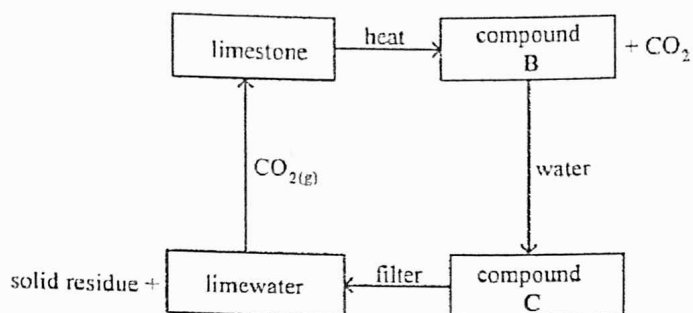


Fig. 8.1

- (i) Write balanced chemical equations for the formation of
1. compound B,
 2. compound C.
- [2]
- (ii) Identify the solid residue.
- [1]
- (iii) State any **one** use of limestone.
- [1]
- (iv) Explain the use of lime in agriculture.
- [1]
9. (a) Table 9.1 is an incomplete table listing some of the gaseous environmental pollutants and their sources.

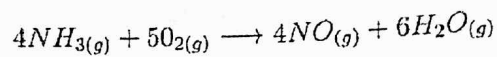
Table 9.1

pollutant	source
methane	fermentation of organic waste
carbon dioxide	car exhaust fumes
nitrogen oxides	car exhaust fumes

- (i) State any other source of nitrogen oxides.
- [1]

- (ii) Name **one** other gaseous environmental pollutant and describe how this pollutant is produced. [2]
- (iii) Describe the environmental effects of the pollutants listed in Table 9.1. [3]
- (b) Describe how pollution from car exhaust fumes can be controlled. [3]
- (c) Unburnt hydrocarbons from car engines can also react with nitrogen oxides as shown in the equation:
- $$x\text{C}_6\text{H}_{14(g)} + y\text{NO}_{2(g)} \longrightarrow z\text{CO}_{2(g)} + 14\text{H}_2\text{O}_{(g)} + \frac{19}{2}\text{N}_{2(g)}$$
- (i) Find the values of
1. x ,
 2. y ,
 3. z .
- [3]
- (ii) Identify, with reasons, the reducing agent in the reaction. [2]
- (iii) State the importance of the reaction in (c) (i). [1]
10. (a) Hydrogen and iodine react according to the following equilibrium reaction:
- $$\text{H}_{2(g)} + \text{I}_{2(g)} \rightleftharpoons 2\text{HI}_{(g)}$$
- (i) State the meaning of the symbol $\rightleftharpoons$. [1]
- (ii) Describe and explain the effect of increasing
1. pressure,
 2. concentration of iodine.
- [4]
- (iii) Describe how the rates of the forward and backward reactions compare at equilibrium. [1]

- (b) The equation for the catalytic oxidation of ammonia is:



- (i) Give **three** suitable conditions for the reaction. [3]
- (ii) Nitric oxide is oxidised in a reversible reaction to nitrogen dioxide.
Write a balanced chemical equation for the reaction. [3]
- (iii) State any **two** uses of nitric acid. [2]
- (c) Describe **one** harmful effect on the environment associated with nitric acid production. [1]

[Turn over

Group																		
I	II											III	IV	V	VI	VII	O	
7 Li Lithium	9 Be Beryllium											1 H Hydrogen	11 B Boron	12 C Carbon	14 N Nitrogen	16 O Oxygen	19 F Fluorine	20 Ne Neon
3 Na Sodium	4 Mg Magnesium											2 He Helium	5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium											13 Al Aluminum	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon	
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton	
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon	
133 Cs Cesium	137 Ba Barium	139 La Lanthanum	140 Hf Hafnium	181 Ta Tantalum	184 W Tungsten	186 Re Rhenium	190 Os Osmium	192 Ir Iridium	195 Pt Platinum	197 Au Gold	201 Hg Mercury	204 Tl Thallium	207 Pb Lead	209 Bi Bismuth	210 Po Polonium	210 At Astatine	222 Rn Radon	
Fr Francium	Ra Radium	Ac Actinium																
87	88	89																

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

Y

X = atomic symbol

5

b = proton (atomic) Number

b = proton (atomic) Number

The volume of one mole of any gas is 28 dm^3 at room temperature and pressure (r.t.p.)